

tems. Phase I of the pilot project is in development with the aim to establish a two-sided energy market.

Another initiative is by power utility company BSES Rajdhani Power Ltd, which plans to use a blockchain-powered ledger for a New-Delhi-based pilot project for P2P solar power trading.

Blockchain can serve as a boon for electric vehicles (EVs), too, through

decentralisation. Besides doing away with the need for a centralised EV charging infrastructure, it eases trading between consumers and charging stations or transport providers.

Yet, to be successful on a large scale, several issues need to be solved. With P2P trading, many new payment platforms for pre-paid meters and pay-as-you-go solutions may come up. Blockchain eliminates the need for

a third party, which challenges traditional market roles like those of metering point operators and suppliers.

Proper governance is necessary to ensure smart management of energy needs and maintain power system stability. Smart grids need user information, which also requires strict protocols to maintain user anonymity and ensure privacy.

—Ayushee Sharma

3 A NEW WORLD OF POSSIBILITIES WITH REALITY 4.0

Extended reality (XR) tools must be developed and deployed in a responsible way for them to become part of daily lives in homes and offices

Many sci-fi movies and shows in the past like *Iron Man* and *Star Trek* have depicted a world of possibilities with technological advancements. But thanks to extended reality (XR), what was once fantasy is now becoming a reality. XR is a broad term that includes augmented reality (AR), virtual reality (VR), mixed reality (MR) and any other immersive technology in the spectrum that merges the real world with a virtual environment.

Anyone intending to use any form of XR in their application first identifies the features or use-cases for which XR can enhance the end-user experience. XR usually requires optic mechanical designs with high precision to create an immersive experience. AR requires superimposition of virtual images over real-world objects using visual markers, location or Simultaneous Localization and Mapping (SLAM) technique. The resulting illusion can be viewed from a smartphone camera or other such devices as smart glasses.

VR, on the other hand, uses a headset to put up a screen in front of the user's eyes to create a simulated 3D environment that provides complete isolation from the real world. Sound effects consistent with visual graphics further engage the user.

MR is an amalgamation of both AR and VR to get the best results.

While growth of XR market started

with the gaming and entertainment industry, it is now being implemented across various fields including engineering, education, healthcare, marketing, automobile, real estate and so on. Highly popular game Pokémon Go, which was introduced in 2016, uses AR to overlay digital creatures (Pokémon) into the real world. Several VR headsets such as Google Cardboard, HTC Vive, GearVR and Oculus Rift are already available in the market.

Even field-of-vision (FOV) is no longer a limitation for the latest devices. One example is startup Varjo, which has recently developed XR-1 MR headset that projects a mix of headset's cameras and 3D images to help customers create high-end 3D designs or conduct employee training.

Benefits of XR solutions are multifold. Besides improving customer experience, it can decrease costs, and increase revenue and productivity of businesses in the long run. One such company creating immersive experiences for enhanced user engagement through XR is Digital Jalebi. The company designs and develops interactive installations and software for spaces like museums, exhibitions and planetariums. Saumik Trivedi, business head, Digital Jalebi, Mumbai, says, "Just like physical and social media presence help customers feel more connected to a brand, having an experiential marketing strategy helps

them talk to the brand and make them feel special, unlike static ads that offer only one-way communication."

Combined with artificial intelligence (AI) and machine learning (ML), XR has significantly impacted the Industry 4.0 revolution. And all this is possible when the technology is still in its infancy. According to a recent report from Research and Markets, global XR market is expected to grow at a CAGR of over 65 per cent from 2019 to 2024. This means it might become common in the future to buy something from a shop or attend a class without being physically present there.

With the introduction of 5G, expectations are even higher. Trivedi says, "5G will help stream high-quality content at very low latency. This is very helpful for VR and 360° content that is generally very heavy compared to 2D content. This will help increase the adoption and viewership of AR and VR content."

Regarding adoption into mainstream, he says, "Key challenges right now are on the hardware front—making it affordable, improving quality of visuals, haptic feedback and improving level of comfort so that content can be consumed for a longer duration." This is necessary to ensure that user health is not affected. Devices require high processing power for processing huge amounts of collected data. This also threatens user privacy as it captures more intimate data like people's